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The impact of public spending on health: does money matter?

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Abstract

We use cross-national data to examine the impact of both public spending on health and non-health factors (economic, educational, cultural) in determining child (under-5) and infant mortality. There are two striking findings. First, the impact of public spending on health is quite small, with a coefficient that is typically both numerically small and statistically insignificant at conventional levels. Independent variation in public spending explains less than one-seventh of 1% of the observed differences in mortality across countries. The estimates imply that for a developing country at average income levels the actual public spending per child death averted is \$50,000–100,000. This stands in marked contrast to the typical range of estimates of the cost effectiveness of medical interventions to avert the largest causes of child mortality in developing countries, which is \$10–4000. We outline three possible explanations for this divergence of the actual and apparent potential of public spending. Second, whereas health spending is not a powerful determinant of mortality, 95% of cross-national variation in mortality can be explained by a country's income per capita, inequality of income distribution, extent of female education, level of ethnic fragmentation, and predominant religion. © 1999 Published by Elsevier Science Ltd. All rights reserved.

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Introduction

In 1995 over nine million children, under five years, in developing countries died avoidable deaths: more than the entire population of Sweden or Zambia.¹ The cumulative human suffering in the individual and familial tragedies behind these statistics is overwhelming and creates a powerful impetus to action. Towards the aim of supporting effective action we examine the determinants of these differences in the widest and best measured indicators of health status: child (under-5 years) and infant mortality. We establish two major points about the cross-national relationship between health status and public spending on health.²

First, there is an enormous gap between the apparent potential of public spending to improve health sta-

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¹ 'Avoidable' deaths are simply defined as the excess of the average death rate for the 0–5 years age group in the low- and middle-income countries 88 per 1000 vs the level in the high-income countries, 9 per 1000. Using a similar approach Gwatkin (1980) calculated the total number of deaths of under-5s to be about 15 million.

² We use the perhaps awkward phrase 'public spending on health' throughout to avoid the ambiguity in the phrase 'public health expenditures' which could mean either 'expenditures on those items classified as "public health" ' or 'all health expenditures by the public sector'.

tus and the actual performance. Reviews of the cost effectiveness of preventive and primary curative interventions suggest that a significant fraction of under five deaths could be avoided for as little as \$10, and in many cases under \$1000, per death averted (Jamison et al., 1993). However, in practice, cross-national differences in public spending on health account for essentially *none* (one seventh of 1%) of the differences in health status. This extremely small actual association estimated from the cross-national data implies that the typical public spending on health per child death averted in developing countries is \$50,000 to 100,000. This is a striking discrepancy between the apparent potential and actual performance.

Second, while public spending appears to explain little, the differences across countries in infant and child mortality are well explained by economic and social factors. The finding that ‘development’ is strongly associated with improvements in mortality is neither surprising nor new (Caldwell, 1986; World Bank, 1993). What is perhaps surprising is the strength of the relationship as essentially all (95%) of the cross-national variations in either under-5 or infant mortality can be

explained by just five factors: the average level of income, the distribution of income, the extent of female education, the extent of ethno-linguistic differences within a country, and whether a country is predominately Muslim. While there are poor countries with exceptionally good health status, properly accounting for income and other economic determinants leaves little to be explained by independent variations in health policy.

Methods: variables, data and procedures

Our empirical strategy is straightforward: we specify and estimate a multivariate regression that explains country level health outcomes with socioeconomic characteristics and public expenditures on health.

We estimate the following equation derived from an aggregate ‘health production function’ that assumes that health outcomes depend on a country’s income, knowledge, and social capability (see Appendix A):

$$\ln(M_i) = \beta_1 \ln(GDP_i/N_i) + \beta_2 \ln(H_i/GDP_i) + \beta_3 X_i + \epsilon_i \quad (1)$$

This equation relates the (natural) log of child (or infant) mortality M , to the log of mean per capita income, GDP/N , the log of the share public spending as a fraction of GDP, H/GDP , the level of female education and a variety of other socio-economic variables (detailed below) X .

Main variables included in the model: Income and public spending on health

The inclusion of income is a necessary control for the universally acknowledged impact of income on health which works through a variety of indirect channels (e.g., better nutrition, better housing, better sanitation).³ The transformation of the income variable to logs achieves two things. First, every study that has examined the issue has shown a non-linear relationship between mortality and income and over the range of mortality outcomes we examine, the non-linearity is adequately captured by a log transformation.^{4,5} Second, this convenient transformation allows comparisons with earlier results as the regression results provide elasticities. (An elasticity reports the percentage change in one variable for a 1% change in the other variable, and is a useful way to compare empirical results as it is scale neutral). The range of ‘elasticities’ measured from cross-national data is between -0.4 and -0.8 , that is a 1% difference in income is associated with a fall of mortality of between 0.4 and 0.8%.⁶

Note that estimating the log of the share of public spending on health in GDP gives numerically equivalent estimates on public spending as estimating the log

³ We do not attempt the exercise of forcing the data to choose between a variety of other variables which are highly related to income through some mechanical statistical procedure like step-wise regression, as was carried out in Hertz et al. (1994).

⁴ Anand and Ravallion’s (1993) argument that average income matters for health only through its effect on the share of the population in poverty is at times misinterpreted as suggesting that ‘income does not matter’. In fact, their finding is not about whether income affects health status but about the appropriate non-linear functional form for the specification for estimating the income effect. They suggest a kink in the relationship whereas we suppose a smooth curve (logarithmic).

⁵ Some have proposed the use of a logistic transformation since there is a physiologically determined lower bound on mortality (Kakwani, 1993) from unavoidable peri-natal deaths but for the developing countries in our sample this lower bound is never approached.

⁶ Kakwani (1993) uses functional forms that allow for varying income elasticity in cross-national data and finds a range of elasticities between -0.5 and -0.6 . Pritchett and Summers (1996) use time series on changes in income and under-5 mortality from 1960 to 1985 and find the long-run elasticity to be between -0.43 and -0.76 (depending on the instruments used in the instrumental variables estimation). Pritchett (1997) uses time-series cross sector panel of data for 22 countries with data going back to 1870 to do fixed effects estimation and finds an infant mortality elasticity with respect to income of -0.59 . Jamison, Wang, Hill and Londono (1996) combine cross section and time series data and find an income elasticity of child mortality of -0.40 in 1960 but of -0.65 in 1990.

of total public spending (as simple arithmetic shows since the log of a ratio is the difference of logs). The only question then is between specifying a functional form such that public spending enters as an absolute level (which ignores the very good reasons for a non-linear relationship between health and spending) or entering public spending as the log of its share in GDP. The latter approach does make for a complicated interpretation of the income term as income enters the equation twice, once in logs and once in logs as the denominator of a ratio. We nevertheless opt for the latter as this yields a straightforward interpretation of public spending on health, our primary variable of interest, and is justified by the analytic framework (see Appendix B).

Choosing the set of other socioeconomic variables

In addition to income and public spending, health outcomes will be affected by a country's level of knowledge and social capabilities, which we allow for in the model by including a set of additional socioeconomic characteristics.

Both aggregate and household level studies show that higher levels of female education are associated with better health status (Hill and King, 1992; Subbarao and Raney, 1995; Caldwell, 1986, 1990; Hobcraft, 1993). Table 1 presents the differences in under-5 mortality by mother's educational status as derived from 45 Demographic and Health Surveys. Mothers who have secondary schooling have child mortality rates 36% lower than those with only primary schooling.

In addition to the level of income, empirical and theoretical considerations suggest greater inequality in income distribution should be associated with worse health status for a given level of income. Flegg (1982) finds that, controlling for female illiteracy, the elasticity of infant mortality with respect to the Gini coefficient, which measures the degree of income inequality, is 0.77.⁷ Using a different approach, Bidani and Ravallion (1997) estimate the relationship between the level (not the log) of life expectancy and infant mortality on a non-linear functional form for income that allows different elasticities for the poor and the non-poor and find strong confirmation of differential impacts.

In addition to the empirical evidence, there is a simple, but not widely recognized, methodological point that with a non-linear specification (e.g., logs) using aggregate data implies that an income inequality

⁷ The Gini coefficient measures the proportional deviation of income distribution from perfect equality, so that a Gini of 0 would be equality and a Gini of 1 would mean one person received all the income. In general, Gini coefficients range from around 0.23 in the former Socialist countries to around 0.6 in the highest inequality countries.

Table 1
Mean (SD) under-5 mortality by education level of the mother: DHS results^a

	Mothers with no schooling	Mothers with primary schooling	Change (percentage relative to none)	Mothers with secondary schooling	Change (percentage relative to primary)	Number of countries
East Asia and Pacific	119.0 (40.3)	69.9 (18.1)	-41.3	39.2 (16.0)	-43.9	3
Latin America and Caribbean	110.9 (52.5)	79.3 (43.1)	-28.5	49.4 (26.1)	-37.7	10
Middle East and North Africa	94.5 (34.3)	62.1 (14.7)	-34.3	41.4 (15.6)	-33.3	6
South Asia	127.7 (41.5)	84.8 (29.9)	-33.6	70.9 (22.9)	-16.4	4
Sub-Saharan Africa	179.8 (60.8)	139.9 (43.0)	-22.2	87.5 (24.0)	-37.5	22
All	144.4 (62.8)	106.5 (49.9)	-26.2	68.4 (30.1)	-35.8	45

^a Compiled from a set of DHS Final Reports from 1987–1995. Where there was a choice, the value for 'completed primary' and/or 'completed secondary' was chosen. In other cases the group may include mothers who have attended but not completed the educational level.

term must be included (Heerink and Folmer, 1994). Our estimates are typical of cross-national results in using the log of average country income, but the log of average income is not the same as the average of log incomes and the difference between these two grows as the variance of income grows.⁸ If the non-linear functional form relating incomes and health that is assumed for cross-national comparisons is assumed the same as that at the individual level within countries, this implies that a measure of the shape of the distribution should *necessarily* be included in the regression, or else it is mis-specified. We include the measured Gini coefficient to capture inequality.

Caldwell (1986) finds that none of the countries with exceptionally good infant mortality performance (relative to income) are predominantly Islamic, whereas many of the poor performers are. We therefore include a dummy variable equal to one if the predominant religion is Muslim and hypothesize that these countries will have higher mortality.

In addition, countries with greater social and political fissures have been shown to have worse economic performance (Easterly and Levine, 1996) and we hypothesize that a measure of ethno-linguistic diversity (defined as the probability two citizens speak the same native language) might also be related to health status. It is well known that in many instances, although certainly not all, ethno-linguistic minorities have worse socioeconomic outcomes than the majority group. With a disadvantaged minority, the larger the group the higher the fractionalization and the higher the average mortality levels. Moreover, there is accumulating evidence that political fractionalization of all kinds, including that occasioned by ethnic conflict, makes it more difficult to achieve desirable outcomes in the provision of public goods (Alesina et al., 1997). Last, we

suspect that urbanization, location in a tropical climate (due to diseases endemic to those regions), and the population's access to safe water would be independently related to health status.

Data

The child mortality data are from UNICEF (1992). While there are difficulties with measuring child mortality, it is arguably superior to alternative measures as a general measure of health status. Life expectancy is not reliably measured in many countries as much of the data reported in official sources are actually extrapolations using child mortality and assumptions about countries' characteristic life tables (e.g., 'North' or 'South' models). Measures of health status that include morbidity (such as Quality Adjusted Life Years or Disability Adjusted Life Years) are more comprehensive, but the data are even less solidly based for cross-national comparisons. We also report results using infant mortality, however it is not our preferred outcome because as infant mortality reaches moderate to low levels it is dominated by peri-natal mortality and hence fails to capture mortality from many of the health conditions of concern which are potentially responsive to health care, such as diarrhea and respiratory infections.

Data on country level health expenditures are a recently compiled data set, based primarily on country sources, such as ministries of health. These data are the latest available update of those which appear in World Bank (1993) and were prepared for The World Bank (1997a). For all their problems of comparability, the data are the best available cross-nationally comparable estimates of total government spending on health. The full list of data sources is in Appendix B.

While this describes our primary data set, we verify that our results are not dependent on the particular data set used. We reproduce our analysis on a completely different data set constructed directly from the World Bank's Social Indicators of Development 1997 database (World Bank, 1997b).

Statistical procedures

Our first approach is to estimate Eq. 1 using Ordinary Least Squares regression (OLS). However, to address the issues of robustness, measurement error, and reverse causation, we use additional econometric procedures.

In order to ensure that the results are not merely driven by a few atypical 'outlying' countries, we undertook two robustness checks. First, to avoid the impact of these influential observations, the two country observations with the largest impact on the parameter vector are dropped. Second, we used median regression, an estimator which is much less sensitive to influential observations than OLS.⁹

⁸ Mathematically, if a function, f , is monotonically increasing ($f' > 0$) but concave function ($f'' < 0$) Jensen's inequality implies that $f(\Sigma x/N) \neq \Sigma f(x)/N$ but there is some value x^* such that $f(x^*) = \Sigma f(x)/N$. The difference between these values depends on the distribution of x , $g(x)$, and mean preserving spreads in $g(x)$ increase the distance between the two. In the particular case of using the log of income, the second order Taylor series expansion gives the difference between the average of log incomes and the log of average incomes as: $\Sigma (y)IN = I(\Sigma y/N) + \sigma_y^2/2(\Sigma/N)^2 + R$. In fact, the difference between the average of the logs and the log of the averages is sometimes defined as an index of inequality.

⁹ Median regression is equivalent to choosing the parameters in order to minimize the sum of absolute deviations (as opposed to the sum of squared deviations as is the case in OLS). Median regression is less sensitive than OLS to influential observations as any deviation from the regression line is 'weighted' by the absolute value of the deviation, not the square of the deviation.

Reverse causation is a potentially serious source of bias in estimating the impact of public sector expenditures as it would tend to reduce the estimated effect of health spending. If the governments of countries that would have higher under-5 mortality devote larger amounts to public spending on health in order to reduce mortality, then the observed regression coefficient might be small, or even negative, and still be consistent with a powerful positive impact of public spending. Measurement error is another potentially serious problem with health spending because the accounting systems which track public spending on health are not coordinated across countries and therefore different countries are measuring different concepts.

For these reasons, we also use an Instrumental Variables (IV) estimation procedure, implemented using Two-Stage Least Squares (2SLS), which can solve both of problems of reverse causality and measurement error. Instrumental variables estimation is an econometric procedure which uses the variation in a set of exogenous variables—or ‘instruments’—to identify the effect of exogenous variations in a variable on the outcome of interest. The method relies on the assumption that the instruments are related to the outcome only through their effect on the variable that is being instrumented.

In this application, a valid instrument for expenditures would need to satisfy three conditions. First, it would need to be correlated with a country's public sector health spending. Second, the measurement error in the instruments would need to be uncorrelated with the measurement error in public sector spending on health. Third, the instrument should not be correlated with under-5 mortality, except through spending. We

use as instruments the average public sector health spending as a share of GDP, and the average defense spending as a share of GDP of a country's geographic neighbors.

Results

Both the direction and magnitude of the estimates on the socio-economic variables are consistent with results reported elsewhere. The elasticity of child mortality with respect to income of around -0.6 (the estimates vary between -0.51 and -0.61) is consistent with previous findings using cross-sectional or time series national level data (see earlier Discussion).¹⁰

The estimates imply roughly 10% lower mortality per additional year of female schooling. This suggests that having four more years of female education than the current average (5 years) is associated with 39% lower under-5 mortality. This is remarkably consistent with the 36% lower under-5 mortality among secondary educated mothers as compared to primary educated mothers estimated from the large collection of household data sets reported in Table 1.

Our results suggest that at the mean level of inequality, a 1% increase in inequality is associated with a 0.5% increase in mortality. This implies that a country with the high inequality of Brazil (Gini of 0.60) compared to that of Sri Lanka (Gini of 0.30) could expect mortality to be 38% higher. The probability that two citizens speak a different native language is associated with higher mortality, a result which is new to the literature. Moving from the very low level of diversity in Costa Rica (0.07) to that of Bolivia (0.70) is associated with a rise in mortality of about 40%. The coefficient on ‘predominantly Muslim’ is very strong for child mortality, implying a child mortality rate higher by 45%, however it is insignificantly associated with infant mortality. This pattern of higher child but not infant mortality is consistent with the pattern of much higher mortality for girls aged 1–4 years (but not aged under 1) in some Muslim countries such as Pakistan and Egypt (Filmer et al., 1998b). Some variables thought to be important, such as the percent of the population that is urban, whether the country is ‘tropical’ (where populations are more exposed to certain diseases) and most surprising, the percent of the population with access to safe water are not found to have significant explanatory power for under-5 mortality.

In addition to their intrinsic interest, these results are reassuring that this specification and methodology are capable of producing reasonable results for a range of variables. This is a ‘state of the art’ equation in terms of ‘controls’ that allows us to confidently examine the impact of spending.

¹⁰ In interpreting estimates of the impact of income on health there is a potentially serious econometric problem of reverse causation, as better health status might cause higher average income. While it is almost certainly true that better health leads to higher income at the individual level (Strauss and Thomas, 1995), the magnitude of the effect is less clear at the aggregate level. Moreover, it is unlikely that the mortality rate of children under five would effect income contemporaneously. Pritchett and Summers (1996), using instrumental variables and fixed effects estimation on a panel of data show that wealthier is causally healthier for the cases of infant mortality, under-5 mortality, and life expectancy. While not going to the same level of detail in the estimation as that study, we use two-stage least squares estimation (column 3 of Table 2) using as instruments for income whether or not a country's primary export is oil, and the percentage of years since 1776 that a country has been independent. Other than the much larger standard errors, the results are largely unaffected, suggesting that whatever role health might have in causing higher incomes it does not affect the estimation of the cross-national impact of income on child mortality.

Table 2
Relationship between mortality and public expenditures on health, controlling for non-health policy factors (1990)^a
?Under-5 mortality rate (ln)

Dependent variable						
Column	1	2	3	4	5	6
Method	OLS	Median ^d regression	Two-stage least squares	Two-stage least squares	OLS	OLS
GDP per capita (ln)	–0.611 (9.71)	–0.570 (4.58)	–0.596 ^c (3.67)	–0.511 ^c (3.39)	–0.598 (9.28)	–0.412 (12.0)
Public expenditures on health (ln of share of GDP)	–0.135 (1.78)	–0.090 (0.677)	–0.192 ^c (0.742)	–0.078 ^c (0.243)	–0.073 (0.92)	
Female education	–0.093 (3.54)	–0.076 (1.65)	–0.091 (2.90)	–0.061 (1.63)	–0.102 (4.03)	–0.008 (3.9)
Income inequality	0.008 (1.28)	0.010 (0.911)	0.008 (1.17)	0.004 (0.603)	0.012 (2.26)	0.14 (2.8)
Predominantly Muslim	0.450 (3.18)	0.265 (0.803)	0.446 (2.91)	0.104 (0.644)	0.408 (3.13)	
Ethno-linguistic fractionalization	0.549 (3.43)	0.303 (1.04)	0.534 (2.90)	0.343 (1.64)	0.625 (4.14)	
Percentage Urban	0.001 (0.459)	0.001 (0.190)	0.001 (0.219)	0.001 (0.359)	0.003 (1.12)	–0.001 (0.23)
Tropical country	–0.051 (0.549)	–0.038 (0.183)	–0.059 (0.619)	–0.165 (1.42)	–0.006 (0.063)	
Access to safe water	–0.001 (0.606)	–0.001 (0.140)	–0.001 (0.390)	–0.003 (0.753)	–0.003 (1.16)	–0.001 (0.23)
Additional variables	Dummy variables for region and a constant term. Dummy variables for when female education, income inequality, ethno-linguistic fractionalization, or access to safe water are missing (in these cases the variable itself is set to zero).					
R-Squared	0.9469	0.7839 ^b	0.9465	0.9361	0.9454	0.923
Number of obs.	98	100	98	98	104	119

^a Countries with largest influence on parameter vector in OLS case and hence dropped are for columns (1) and (3) Zaire and Korea (South), for column (4) Israel and Turkey, for column (5) Rwanda and Korea (South).

^b The *R*-Squared in column 2 is the pseudo *R*-Squared calculated as 1 minus the ratio of the sum of absolute deviations from the predicted median to those from the actual median.

^c Instruments are, neighbors public health spending, neighbors military spending, whether or not the country's main export is oil, years since 1776 that the country has been independent.

^d *T*-statistics derived from bootstrapping with 100 iterations.

Impact of public spending on health

How large is the impact of public spending on health?¹¹ The OLS regression in column 1 of Table 2 shows higher public expenditures on health do appear to be associated with lower under-5 mortality. But the association is both empirically small (with an elasticity of -0.135) and imprecisely estimated: the effect is statistically significant only at the 10% level (with a t -statistic of 1.78).

How robust is this result? There are four indications that this finding is not an artifact of data or

¹¹ There are three reasons we do not estimate health status as a function of private expenditures on health. First, when private expenditure decisions are voluntarily made then, in choosing to spend, people at least believe they will receive value for money. This active choice makes private expenditures certainly, and perhaps irredeemably, endogenous and it is, very difficult to separate out the impact of expenditures on health status from health status on expenditures. Second, the best the data can give is total expenditures on health, but cannot decompose total expenditures into prices and quantities so there is no way of identifying the impact of increased quantities of privately purchased health services. Therefore any cross-national regression between private expenditures and health status has no obvious interpretation. Third, while public expenditures are a policy choice, private expenditures are an outcome, influenced, but not determined by policy.

¹² The smaller income elasticity (-0.412 vs -0.596) is partly explained by the fact that non purchasing-power-adjusted estimates systematically underestimate the incomes of poorer countries which would bias the estimate towards zero. In addition, random measurement error in long-run income estimates due to exchange rate fluctuations may also cause a bias towards zero. When using purchasing-power-adjusted GDP per capita (from the Penn World Tables database) as an instrument for GNP per capita with the set of countries included in our main analysis, over 92% of the variation is explained, and the income coefficient rises (in absolute value) to -0.49 .

¹³ The incremental R^2 is generally not appropriate as a statistical procedure for resolving questions about the relative importance of variables, but if one has prior non-statistical knowledge about the causal ordering among the independent variables it is appropriate. We resolve the problem of attribution of effects in a multivariate regression by having included only non-health factors that may cause higher health spending but which are not directly affected by higher health spending. That is, we assume more female education or lower inequality might lead to higher health spending but not vice versa. This means we attribute to female education, for example, both whatever health improvements it may cause directly through behavioral changes and whatever indirect impact it may have through more spending. This assumption about the causal ordering of health spending and non-health factors is obvious for nearly all of the variables. The only problematic case is of the joint determination of health status and income which we address using instrumental variables.

method. First, the estimated impact of health spending using median regression, presented in column 2 of Table 2 is similar, although slightly smaller in absolute terms: -0.09 vs -0.135 . The biggest change is in the statistical significance of the estimates, the t -statistic is only 0.67 vs 1.78, although this lower precision is not surprising as OLS is the more efficient estimator.

Second, the two-stage least squares (2SLS) results, presented in column 3 of Table 2, are similar to the OLS results. The coefficient on public sector health spending is approximately 40% larger (in absolute terms) -0.192 vs -0.135 . This change in magnitude is consistent with attenuation bias due to measurement error in the OLS estimates. However, the 2SLS coefficient is much more imprecisely measured and is statistically insignificantly different from zero (with a t -statistic of only 0.74). This is a typical 'good news–bad news' result from instrumental variables estimation. Those who would want to argue the effect of public spending is understated due to measurement error have a higher point estimate to support their case. On the other hand, skeptics can point to the imprecision of the estimate and hence the very low significance level as not constituting convincing evidence of a powerful impact.

Third, when using infant mortality as the outcome variable (and with our preferred 2SLS estimation) the results are again similar, although the coefficient on public sector health spending is even smaller in absolute terms (-0.078 vs -0.192), and statistically insignificantly different from zero (with a t -statistic of 0.92). This is consistent with the higher 'genetic' component of neo-natal deaths which are the largest component of infant mortality (Rutstein, 1984).

Fourth, the results using the alternative data set, reported in column 6 of Table 2, again show a small and insignificant point estimate for public spending on health. The estimated elasticity is only -0.073 (t -statistic of 0.92) for the 119 countries for which the relationship can be estimated.¹²

The small impact of health spending translates into very low explanatory power. Comparing column 1 with column 5 of Table 2 in which spending is not included shows that public spending on health explains very little of the variation in under-5 mortality over and above that which can be explained by non health-sector variables: the incremental R^2 is equal to 0.0015. This means that only approximately one-seventh of 1% of all mortality variations are explained by differences in spending. Introducing public sector health expenditures does not affect the other coefficients in the regression by much even though it is positively correlated with some, particularly GDP per capita.¹³ Moreover, the proportion of the differences in the

under-5 mortality rate across 104 countries explained by income, education and social factors is 94%.¹⁴

These regressions answer a simple question: do the star performers in health—countries that do well in health performance for their income, education, and social factors—have greater public spending on health? The regression results can be intuitively understood by examining the difference in public spending on health between the ‘10 best’ and the ‘10 worst’ health performers conditional on socio-economic factors. This difference is very small, while the differences in public spending among countries within the ‘10 best’ or ‘10 worst’ groups are very large. The average fraction of GDP devoted to public spending on health among the ‘10 best’ is 2.05%, vs an only slightly lower average of 1.82% among the ‘10 worst’ and when comparing medians the ‘worst’ spend more, 1.59 vs 1.35 (Table 3).

Jamaica, with mortality 80% better than predicted spends 2.89% of GDP while Brazil, with mortality almost 50% worse than predicted, spends 2.97%. Sri Lanka, the second best health performer, spends 1.67% compared to one of the worst, South Korea,

which spends 1.88%. These are stark examples of the similarities of spending in countries with very different health performance.

Within the ‘10 best’ and ‘10 worst’ groups there are large variations in the public sector health spending as a share of GDP. Costa Rica does well and has public spending of 7.49% of GDP while Jamaica and Sri Lanka do well spending only 2.89 and 1.67% of GDP respectively.

Relationship to previous literature

These results on public spending should perhaps not come as a surprise. Empirical studies that include health resource variables, such as physicians, nurses, or hospital beds per capita rarely find large and significant impacts (Kim and Moody, 1992). There have been very few previous estimates of the impact of public spending in developing countries, mostly because of the lack of appropriate data on public expenditures. Musgrove (1996) summarizes the few studies that have been done and concludes that ‘multivariate estimates of the determinants of child mortality give much the same answer: income is always significant, but the health share in GDP, the public share in health spending, and the share of public spending on health in GDP never are’.¹⁵

Three studies claim to find a significant impact of public spending (Anand and Ravallion, 1993; Bidani and Ravallion, 1997; Jamison et al., 1996). Anand and Ravallion (1993), however, warn that ‘[a] number of caveats should be noted about these results. They are based only on the patterns observed in this sample of 22 countries’. As we emphasize below the impact of public spending on health is not an immutable parameter, but is likely to vary widely from country to country and hence the results will be sensitive to the sample used. Bidani and Ravallion (1997) use a particular functional form to separate out the impact of various variables into their impact on the poor and non-poor. They consistently find an impact of public spending for the poor but not for the non-poor, however, their findings are still consistent with a small and insignificant impact of spending on aggregate health status.¹⁶ Moreover, their findings highlight the importance of considering the incidence of the health benefits: they benefit (at best) only the poor so cuts without reallocation would fall on the poor. As discussed below, the impact on the poor vs the non-poor will also not be a constant or immutable parameter, but will depend on the composition and efficacy of public spending.

¹⁴ The high explanatory power of these regressions is even more impressive when one considers the role of measurement error, both in the dependent variables of mortality rates and in the independent variables. Indeed, many researchers do not trust cross-national comparisons because they doubt the data are sufficiently reliable (Srinivasan, 1994). While it is true that there are difficulties in accurately measuring child mortality, incomes, and educational levels across countries, this cannot be an explanation for a high degree of explanatory power; the more strongly one believes the data is ‘bad’ the more puzzling is the high explanatory power of the regression. In the case of child mortality a recent compilation of mortality estimates has multiple estimates for the same country for the same year, which can be used to estimate the degree of pure measurement error (United Nations, 1992). The differences across estimates using different surveys and different methods for the same periods are substantial: the four estimates of child mortality per 1000 in Egypt in 1980 are 203, 167, 171 and 97, the four estimates for Ghana in 1975 are 187, 171, 130 and 125. The average coefficient of variation of these different measurements for a sample of countries with repeated measurements is 0.129 which is an estimate of the pure measurement error in the cross-national data, compared to a coefficient of variation for the sample as a whole of 0.880. This degree of measurement error suggests the maximum achievable R^2 , if all the true variation were explained, is 0.978. The effect of measurement error in the independent variables has a similar effect in lowering the feasible R^2 .

¹⁵ There is a larger literature in developed countries, which also tends to find little impact of health spending, with some exceptions (Wolfe, 1986).

¹⁶ For infant mortality they find that the elasticity at the population weighted means is -0.213 for the poor and -0.056 for the non-poor, which is roughly comparable to our estimated elasticity -0.078 in the aggregate.

Table 3

Ten best and 10 worst from residuals of under-5 mortality regression excluding public health expenditures as a share of GDP (Table 2, column 5)^a

	Country	Residual	Under-5 mortality rate	Public health expenditures %GDP
Best 10	Jamaica	−0.795	20	2.896
	Sri Lanka	−0.519	35	1.673
	Costa Rica	−0.463	22	7.494
	Singapore	−0.451	9	0.985
	CAR.	−0.419	169	0.954
	Kenya	−0.416	108	1.647
	Zaire ⁺	−0.375	200	0.180
	Trinidad and Tobago	−0.370	17	2.735
	Ethiopia ⁺⁺	−0.364	220	0.839
	Myanmar ⁺	−0.357	88	1.045
Mean (unweighted)				2.05
Median				1.35
Worst 10	Philippines	0.421	69	0.760
	Mozambique ⁺	0.432	297	4.385
	Bangladesh*	0.443	180	1.072
	Syria ⁺	0.461	59	n/a
	Bolivia	0.461	160	1.599
	Peru	0.502	116	1.200
	Brazil	0.509	83	2.971
	Indonesia	0.634	97	0.571
	Rwanda	0.807	198	1.891
	Korea (South)	1.12	30	1.883
Mean (unweighted)				1.82
Median				1.59

^a Missing values (with dummy variable flags) in the regression for (+) inequality (~) female education (*) ethnolinguistic fractionalization.

Realized cost, potential cost, and interpretation

Overall, these regressions show that differences in public sector spending on health do not go far in explaining why some countries have high, and others low, child mortality. Doubling the share of GDP devoted to public spending on health from the mean

of 2.96 to 5.92% is associated with an improvement in mortality of only between 9 and 13%.

Realized cost

Through some algebraic manipulation, the regression results can be used to derive the cost, in terms

Table 4

Cost of averting a death derived from different specifications of health status regressions^a

Source of estimate of α	Value of α (β)	Increasing public expenditures on health	Transferring expenditures from non-health to health	Increasing non-health GDP
Two-stage least squares	−0.192 (−0.404)	\$47,112	\$49,381	\$1,025,398
OLS	−0.135 (−0.476)	\$66,680	\$72,202	\$871,894
Median regression	−0.090 (−0.481)	\$100,850	\$114,193	\$863,107

^a α is the coefficient on the log of share of public health in GDP, β is the implied coefficient on the share of non-health spending (see Appendix B).

Table 5

Percent of deaths by cause for population subgroups and cost per death averted^a

	Percent of deaths by cause (Estimates for 1985)			Estimates of cost per death averted of treatment Developing countries
	Industrial market economies	Developing countries		
	All ages	All ages	Under-5	
Infectious and parasitic	4.6	44.9	27.7	
Diarrhea diseases	0.0	13.2	10.6	\$1000 to \$10,000 ^b
Tuberculosis	0.2	7.9	0.8	\$20 to \$76 ^c
Acute Respiratory Infection (ARI)	3.6	16.6	11.3	\$379 to \$1610 ^b
Non-ARI measles and whooping cough	0.0	1.8	1.8	\$10 to \$561 ^d
Malaria	0.0	2.6	2.0	\$78 to \$990 ^e
Other	0.0	2.6	1.2	
Complications of pregnancy	0.0	1.3	0.0	\$836 to \$3967 ^f
Prenatal conditions	0.6	8.4	8.4	
Non-communicable	78.6	29.8	0.0	
External (including injuries)	17.0	15.6	2.4	
Total	100	100	38.5	

^a Source: Causes of death derived from Lopez (1993), sources for costs are Diarrhea—Martines et al. (1993), TB—Murray et al. (1993), ARI—Stansfield and Shepard (1993), Measles—Foster et al. (1993), Malaria—Najera et al. (1993).

^b Current US dollars, from model estimates.

^c 1989 US dollars, from three Sub-Saharan African countries (Malawi, Mozambique, and Tanzania).

^d Current US dollars, from model estimates, measles only.

^e Malaria control, 1987 US dollars, from two studies, one in Sri Lanka (\$78) using insecticides, one for Developing countries (\$990) using vector control.

^f Current US dollars, from model estimates, maternal and non-natal deaths are not distinguished in this study. Note that these costs are estimated using different methods (e.g., model estimates vs observed ranges) and should be only used with extreme caution.

of public spending on health, of averting the death of a child under-5 (see Appendix A for details on this derivation).

¹⁷ The calculations are made for a country with an under-5 mortality rate of 126, a population of about 7 million, a GDP per capita of \$1816 (in 1985 international dollars), a crude birth rate of 34.15, and whose public health expenditures are 2.1% of GDP.

¹⁸ Our confidence in these methods for producing reasonable magistrates is bolstered by the fact that Viscusi (1993) reports the estimated individual 'willingness to pay' to avert a statistical death for the US. The range of estimates is very wide, between \$2.5 and \$7.3 million which is between 124 and 375 times the average workers' income. There are few studies like this for developing countries, but a hedonic wage regression for Taiwan, China, gives an estimate of \$650,000 which is 54 times average worker income (assuming GDP per capita is half average worker income). If we view the 'non-public health share' part of our estimates to be related to private expenditures on health, then our estimate from the two-stage least squares estimate is \$1 million, or 265 times the average worker income of the countries under study (again assuming GDP per capita is half average worker income) which is lower than the US range but higher than the Taiwanese estimates, but in both cases within a small multiple.

The results imply that the cost of averting the death of a child under-5 in a typical high-mortality developing country, in terms of public sector health spending, is equal to \$47,112 using the two-stage least squares point estimate. The range of the three different estimation techniques (two-stage least squares, OLS, median regression) is \$47,112 to \$100,927 (Table 4).¹⁷ The amount for non-health spending is between \$863,000 and \$1 million.¹⁸

Are these results about the magnitude of public expenditures required to avert a death surprising or revealing? They do appear to be completely at odds with more conventional health intervention based cost-effectiveness estimates of the minimum required cost to avert a death for most childhood conditions.

Potential cost

The highest fraction of deaths, and in particular of child deaths, in developing countries are due to infectious and parasitic diseases (such as diarrheal diseases, tuberculosis, or malaria). As reported in Table 5, about 28% of all deaths (including those of adults) in developing countries are due to infectious and parasitic diseases among children under-5. The two most promi-

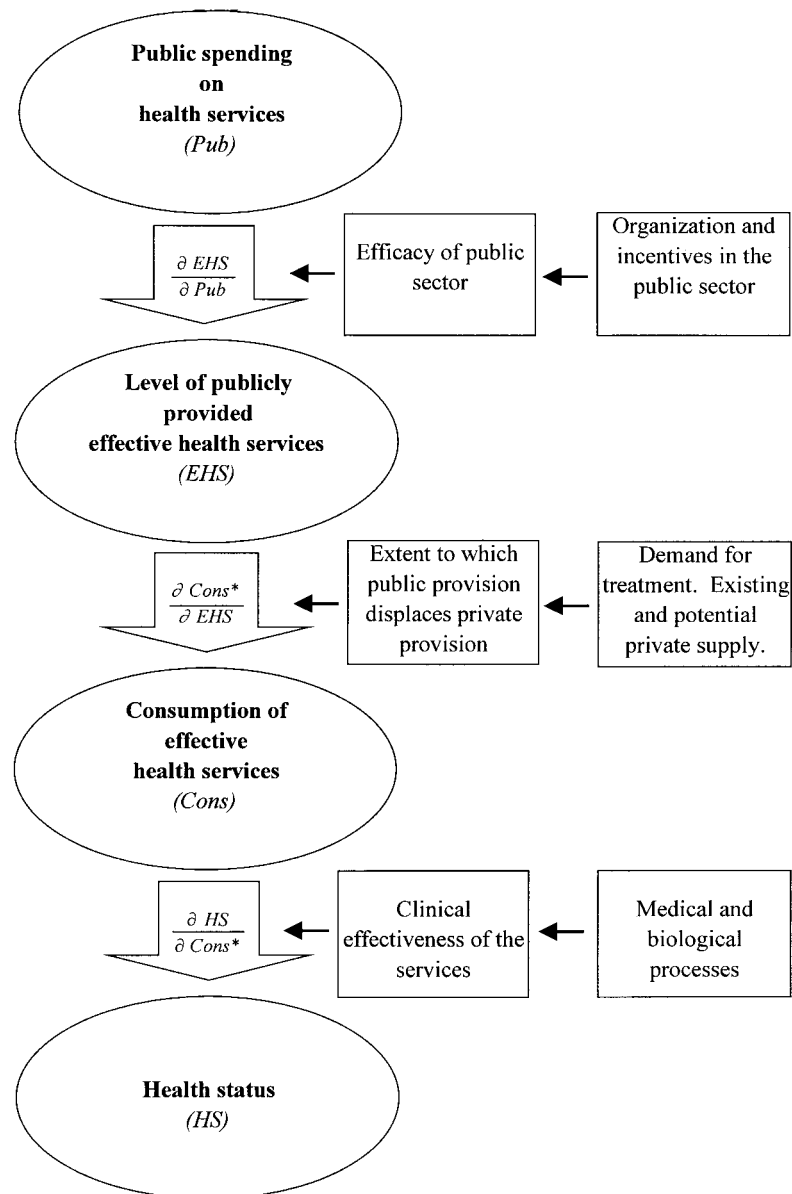


Fig. 1. From public spending to better health.

nent among those are children under-5 dying of diarrheal diseases and Acute Respiratory Infections (ARI) which each account for slightly over 10% of all deaths. An additional 8% of all deaths in developing countries are due to peri-natal conditions. Since these are the largest causes of death, a substantial reduction in overall mortality must involve a fall in mortality from these causes.

A selected group of estimates of the cost per death averted for a variety of treatments that address these conditions is also presented in Table 5 (derived from

Jamison et al., 1993). The ranges for the three biggest killers are \$1000 to \$10,000 for diarrheal diseases, and a much more modest \$379 to \$1610 for ARI, and \$836 to \$3967 for the complications of pregnancy. From these numbers it would appear that public sector intervention to treat each of these conditions could be done relatively cheaply and could save many lives.

The results discussed so far imply that between the 'effectiveness' of public spending on health and the 'cost effectiveness' of available medical interventions

there is a several orders of magnitude gap in the cost per averting a child death. The cross-national regression results suggest a range between about \$50,000 and \$100,000 whereas the medical intervention cost effectiveness figures suggest a range between \$10 and \$4000. While both of these sets of calculations must be considered extremely rough, the differences are roughly 100 fold, which suggest real and important differences.

Interpretation

Understanding why public spending on health has not had a strong effect on reducing mortality is crucial to designing public policy to reduce excess mortality and morbidity in developing countries. We stress that we are not suggesting that medical services are not (potentially) effective. We are definitely not arguing that penicillin, immunizations, or Oral Rehydration Therapy are ineffective as health interventions. But the impact of public spending on health is much more complicated than the effectiveness of the services purchased. There are three possibilities as to why, even though there is the potential for reducing deaths cheaply, it has not happened in practice.

These three can be expressed as a chain from public spending to health status (Fig. 1).¹⁹

- *Health production function.* The change in health status is affected as a proximate matter by changes in the consumption of various health services some of which are more, and some less, effective in improving health.
- *Net public sector impact.* However, even of particular services that are 'cost effective' in improving health, this does not mean that *public spending* on those goods would be cost effective in improving health. Additional consumption in the public sector occasioned by public supply will crowd out, to differing extents, services that would have been consumed anyway. So a second factor is how the consumption of health services (of various kinds) changes with a change in effective public supply of those services.
- *Public sector efficacy.* The impact of public spending will depend first on the degree to which public spending is able to create effective public services. That is, in some countries a dollar spent in the public sector on health will create facilities and services

which are effective at improving health status while in other countries it will create not just expensive services but no services at all.²⁰

The importance of viewing this as a chain is that the total impact depends on all elements. For public spending to improve health cheaply, that is if 'money matters', three things must happen: first, public spending must create effective health services, second the existence of those new public services has to change the total amount of effective health services consumed by the population, and third, the additional services consumed have to be cost-effective in improving health. If any one of these conditions is not met the actual cost effectiveness of services can be very high, even if there existing services with low potential cost per death averted.

This simple framework provides a useful way of summarizing the differing conclusions and policy reform implications of various schools of thought on policy reform. One group focuses on the *cost-effectiveness of public spending* and advocates changes in the composition of public spending towards more cost effective (generally cheaper) services. A second group focuses on the *net impact of public sector supply* and emphasizes the role of government to supplement, not replace, the market for low cost services. The third group focuses on *public sector efficacy* and emphasizes improving the operational performance of public sector health care providers.

Of course these three schools of thought are not mutually exclusive and, as is clear from the framework presented here, all three matter. The hard question for public policy is their relative empirical importance in any given situation. In some cases the government may be quite effective at supplying services and the private market weak (for example in inaccessible, poor rural areas). In others cases the government may be quite ineffective and the private sector strong. In some cases the market for inexpensive curative services (of a Primary Health Care type) may work quite well while elsewhere insurance markets are weak so public sector supply of tertiary services is the right policy.

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¹⁹ The mathematically inclined can see this as the total derivative broken into the chain of partial derivatives.

²⁰ Filmer et al. (1998a) is an analytic review of the theoretical and empirical literature relevant to these three 'links in the chain' leading from public spending to better health.

Appendix A. Regression coefficients and public spending on health per death averted

In order to motivate the relationship between the regression coefficient and public spending on health per death averted, think of the specification of the regressions in Table 1 as being generated from the following simple model of the determination of aggregate health status

$$\text{Health Status} = (H_i/N_i)^\alpha \times (NH_i/N_i)^\beta \times e^{A_i} \quad (2)$$

where H_i is public expenditures in the health sector of country i , NH is the 'rest of GDP' (and includes all non-public sector health spending), N is the population, and A is a country specific factor. Dividing the numerators and denominators through by GDP and taking logs implies that the log of health status is a function of the log of the public health expenditures as a share of GDP, non-public health sector spending as a share of GDP (i.e., with $H + NH = \text{GDP}$), GDP per capita, and the country specific factor, as in

$$\ln(\text{Health Status}_i) = \alpha \ln(H_i/\text{GDP}_i) + \beta \ln(NH_i/\text{GDP}_i) + (\alpha + \beta) \ln(\text{GDP}_i/N_i) + A_i \quad (3)$$

Allowing A_i to depend on the a set of observable and unobservable characteristics leads to the regression estimations reported in Table 2.

Eq. (2) implies an expression for the relationship between the public expenditures on health as a share of GDP and health status, that is

$$\begin{aligned} \partial(\text{Health Status}_i)/\partial(H_i/\text{GDP}_i) \\ = (\text{Health Status}_i)/(\text{GDP}_i/N_i) \times \alpha/(H_i/\text{GDP}_i) \end{aligned} \quad (4)$$

Eq. (4) can be expressed in terms of the number of deaths averted and, in turn, can be inverted to derive the public sector health spending per additional death averted. An equivalent expression can be derived for the amount of GDP that is not public expenditures on health. From the difference one can calculate the amount that would need to be transferred from GDP to public sector health expenditures in order to avert an additional death. The values derived from these calculations, estimated at the median characteristics of countries with under-5 mortality rates greater than 20 are presented in Table 4.

Appendix B. Cross national data availability and definitions, and additional results

This appendix provides details on four aspects of the cross-national regressions: data on health status, data

on independent variables, data on health sector strategy, and functional form.

Data on health status

The data used here are as reported by UNICEF (1992) except for the mortality rates for Zaire which are not believable: under-5 mortality rate for 1990 is reported as 130 and infant mortality rate as 79. These are replaced with the, more reliable, estimates reported in United Nations (1992) for 1984. These are 200 for under-5 mortality and 126 for infant mortality.

Data on non-health sector variables

Income: Real GDP per capita in 1985 international dollars (i.e. adjusted for Purchasing Power Parity) are from the Penn World Tables 5.6.

Education: Average education levels for women over 15 are from Barro and Lee (1996).

Income inequality: Gini coefficient as calculated by Deininger and Squire (1996) multiplied by 100.

Percent urban: Percent of the country's population that lives in urban areas. From the World Bank's Social Indicators of Development database.

Predominantly Muslim: Dummy equal to one if over 90% of the country's population is Muslim.

Ethno-linguistic fractionalization: Measures the probability, in 1960, that two randomly selected people from a given country will not belong to the same ethno-linguistic group as reported in Easterly and Levine (1996).

Tropical country: Dummy equal to one if part of the country's territory lies within 20° of the equator.

Access to safe water: Percent of population with access to safe water. From the World Bank's Social Indicators of Development database.

Oil exporter: Dummy equal to one if the country primary export is fuels (mainly oil) as classified by the World Bank's World Development Indicators (1996) plus Kuwait.

Years independent: The percentage of years since 1776 that a country has been independent, as reported in Easterly and Levine (1996).

Defense spending: Defense spending as a share of GDP, as reported in CIA (1994)

Health Expenditures: As part of the World Development Report 1993, *Investing in Health*, figures on the magnitude of health expenditures were generated (Murray et al., 1993). These have undergone various updates, based on country level reports. We use the latest data available in our analysis (Table A1).

Table A1

Summary statistics for estimation sample (Number of observations = 100)

	Mean	SD	Correlation with GDP per capita (ln)
Basic variables used in the regressions			
Under-5 mortality rate (ln)	3.928	1.183	−0.9274
Under-1 mortality rate (ln)	3.605	1.053	−0.9200
GDP per capita (ln)	7.914	1.151	1.000
Public health expend. (ln of share of GDP)	−3.769	0.7591	0.6228
Female education	4.971	2.753	0.8115
Income inequality	40.93	8.706	−0.2432
Percent urban	49.96	25.01	0.8075
Predominantly Muslim	0.120	0.327	0.0096
Ethno-linguistic fractionalization	0.4219	0.2871	−0.5112
Tropical country	0.620	0.488	−0.6445
Access to safe water	67.13	25.73	0.6909
Regional dummy variables			
East Asia and Pacific	0.10	0.302	0.0377
Latin America and Caribbean	0.21	0.409	0.0090
Middle East and North Africa	0.09	0.288	0.1236
South Asia	0.05	0.219	−0.1346
Sub-Saharan Africa	0.31	0.465	−0.6812
Missing observation flags			
Female education missing	0.11	0.314	−0.2463
Income inequality missing	0.23	0.423	−0.2649
Ethno-linguistic fractionalization missing	0.08	0.273	0.0390
Access to safe water missing	0.08	0.273	0.3405
Raw (un-logged) averages			
Under-5 mortality rate	87.75	77.25	
Under-1 mortality rate	56.96	44.66	
GDP per capita	5004.00	5300.00	
Public health expend. (Share of GDP)	0.02967	0.01998	

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Update

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Erratum

Filmer, D., Pritchett, L., 1999. The impact of public spending on health: does money matter? *Social Science & Medicine* 49, 1309–1323[☆]

Unfortunately when this article was published, Table 2 contained two errors. The first error was that “Under-5 mortality rate (ln)” should have been the heading for the first three columns. The second error

was that the coefficient on “Public expenditures on health (ln share of GDP)” in column five should have appeared in column six and column five should have been blank. The correct table is printed overleaf

[☆] PII of original article: S0277-9536(99)00150-1

Table 2
Relationship between mortality and public expenditures on health, controlling for non-health policy factors (1990)^a

Dependent variable	Under-5 mortality rate (ln)			Infant mortality rate (ln)	Under-5 mortality rate (ln)	World Bank SID data-under-5 mortality
Column	1	2	3	4	5	6
Method	OLS	Median ^d regression	Two-stage least squares	Two-stage least squares	OLS	OLS
GDP per capita (ln)	–0.611 (9.71)	–0.570 (4.58)	–0.596 ^c (3.67)	–0.511 ^c (3.39)	–0.598 (9.28)	–0.412 (12.0)
Public expenditures on health (ln of share of GDP)	–0.135 (1.78)	–0.090 (0.677)	–0.192 ^c (0.742)	–0.078 ^c (0.243)		–0.073 (0.92)
Female education	–0.093 (3.54)	–0.076 (1.65)	–0.091 (2.90)	–0.061 (1.63)	–0.102 (4.03)	–0.008 (3.9)
Income inequality	0.008 (1.28)	0.010 (0.911)	0.008 (1.17)	0.004 (0.603)	0.012 (2.26)	0.14 (2.8)
Predominantly Muslim	0.450 (3.18)	0.265 (0.803)	0.446 (2.91)	0.104 (0.644)	0.408 (3.13)	
Ethno-linguistic fractionalization	0.549 (3.43)	0.303 (1.04)	0.534 (2.90)	0.343 (1.64)	0.625 (4.14)	
Percentage Urban	0.001 (0.459)	0.001 (0.190)	0.001 (0.219)	0.001 (0.359)	0.003 (1.12)	–0.001 (0.23)
Tropical country	–0.051 (0.549)	–0.038 (0.183)	–0.059 (0.619)	–0.165 (1.42)	–0.006 (0.063)	
Access to safe water	–0.001 (0.606)	–0.001 (0.140)	–0.001 (0.390)	–0.003 (0.753)	–0.003 (1.16)	–0.001 (0.23)
Additional variables	Dummy variables for region and a constant term. Dummy variables for when female education, income inequality, ethno-linguistic fractionalization, or access to safe water are missing (in these cases the variable itself is set to zero).					
R^2	0.9469	0.7839 ^b	0.9465	0.9361	0.9454	0.923
Number of obs.	98	100	98	98	104	119

^a Countries with largest influence on parameter vector in OLS case and hence dropped are for columns (1) and (3) Zaire and Korea (South), for column (4) Israel and Turkey, for column (5) Rwanda and Korea (South).

^b The R^2 in column 2 is the pseudo R^2 calculated as 1 minus the ratio of the sum of absolute deviations from the predicted median to those from the actual median.

^c Instruments are, neighbors public health spending, neighbors military spending, whether or not the country's main export is oil, years since 1776 that the country has been independent.

^d T -statistics derived from bootstrapping with 100 iterations.